

Andrew Koh

Postgraduate Engineering Student

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EXPERIENCE

Production and Manufacturing Assistant

Aug. 2022 – Current

Agile Workspace Ltd.

Auckland, New Zealand

- **LLM & Workflow Automation:** Built an end-to-end automated rostering pipeline using a Facebook Messenger chatbot and LLM API, eliminating manual manager entry and reducing scheduling overhead per update to near zero.
- **Embedded Systems Development:** Developed an occupancy sensor application on an ATmega MCU using accelerometer and PIR sensors, achieving an 87% reduction in detection time.
- **Hardware Manufacturing & QA:** Manufacturing, documentation, and integration-testing of diverse electronic hardware including embedded devices, network controllers, and mini PCs over various communication protocols including: Wi-Fi, radio, and Ethernet.

Firmware Engineer Intern

Nov 2023 - Feb 2024

Resideo

Auckland, New Zealand

- **Embedded Driver Development:** Designed and implemented peripheral drivers on an ARM Cortex-M0+ microcontroller to control a motorized camera lens, allowing precise hardware-level actuation.
- **SoC-MCU Communication:** Defined and extended a custom I2C protocol with 6 new lens control commands and enhanced the SoC CLI, enabling end-to-end MCU control through the host interface.
- **Autofocus Algorithm:** Designed a contrast-based autofocus algorithm using a finite state machine, significantly reducing lens traversal distance compared to a full-range sweep.

PROJECTS

Final Year Project — 6D Pose Estimation for Robotic Assembly

Mar 2024 – Oct 2024

- **3D Computer Vision:** Implemented 6D pose estimation algorithms using a depth camera for real-time point cloud acquisition, achieving less than 1mm of accuracy error in on target objects.
- **Motion Control:** Applied forward and inverse kinematics to translate pose estimates into joint-space commands, enabling the robotic arm to position its end-effector to target objects.
- **ROS Integration:** Developed the full pipeline within the ROS framework for sensor fusion and processing.

Underactuated Anthropomorphic Robotic Gripper

Aug 2025 - Oct 2025

- **Rapid Prototyping:** Designed and rapidly prototyped an underactuated anthropomorphic gripper using 3D printing to manufacture phalanges, actuation mechanisms, and compliant padding.
- **Tendon-Driven Actuation:** Developed a single stepper motor-driven tendon system employing Whipple tree linkages to coordinate multi-finger actuation.
- Achieved a maximum grip force of 50 N before mechanical failure using a single Dynamixel actuator.

Fire-Fighting Robot | C++, PID Control, Kalman Filtering, ROS

Apr 2024 - Jun 2024

- **Autonomous Navigation:** Built an autonomous robot that detected and extinguished two fires on opposite sides of a test table in <60 seconds with 0 collisions.
- **Subsumption Architecture:** coordinating obstacle avoidance, fire detection, and extinguishing.
- **PID & Sensor Filtering:** Tuned PID controllers with Kalman and convolution-based sensor filtering, achieving less than 3° orientation accuracy across 0–1.5 m and varying lighting conditions.

EDUCATION

University of Auckland

Bachelor of Engineering (Honours) - Mechatronics, First Class Honours | GPA: 7.7/9

Auckland, New Zealand

2021 - 2024

University of Auckland

Master of Engineering - Robotics and Automation

Auckland, New Zealand

Exp Jul 26 - Jun 27

TECHNICAL SKILLS

Languages: Python, C/C++

Hardware/Architecture: AVR, ARM, RISC-V

Frameworks: ROS, Flask, React, FreeRTOS, Unity Testing, Playwright, Platform IO

Robotics: Control Theory, Rapid Prototyping, Omron PLC

CAD: Inventor, Onshape